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The Convergence Rate of a Multigrid Method with Gauss-Seidel Relaxation for the Poisson Equation

Based on the paper by Braess [1]

1. Setup

We start with the Poisson equation with Dirichlet boundary conditions:

$$\begin{aligned} -\Delta u &= f \qquad \text{in } \Omega \\ u &= 0 \qquad \text{on } \partial\Omega. \end{aligned} \tag{1}$$

The equivalent weak form is

$$a(u, v) = (f, v)_0 \qquad \forall v \in S_h \tag{2}$$

where $a(\cdot, \cdot)$ is the bilinear form

$$a(u, v) := \int_\Omega \nabla u \cdot \nabla v \ \mathrm{d}\xi \ \mathrm{d}\eta = \int_\Omega \left(u_\xi v_\xi + u_\eta v_\eta\right) \ \mathrm{d}\xi \ \mathrm{d}\eta \tag{3}$$

and $(\cdot, \cdot)_0$ is the inner product of L^2,

$$(f, v)_0 := \int_\Omega f v \ \mathrm{d}\xi \ \mathrm{d}\eta. \tag{4}$$

We also introduce the energy functional

$$J(u) := a(u, u) - 2(f, u)_0. \tag{5}$$

It can be shown that the solution of the strong (or weak) form of the Poisson equation is equivalent to finding the minimizer of J.

In this proof we typically assume the homogeneous problem ($f = 0$) since we only want to examine error convergence. In this case, the exact solution is $u^* = 0$. Hence, an approximation u^k is equivalent to the error $e^k = u^k - u^* = u^k$. Furthermore, J simplifies to $J(u) = a(u, u)$.

The domain is discretized using the following grids. The coarse grid Ω_H has the same structure as the fine grid Ω_h, only that it is rotated by 45° and has a spacing of $H = \sqrt{2}h$

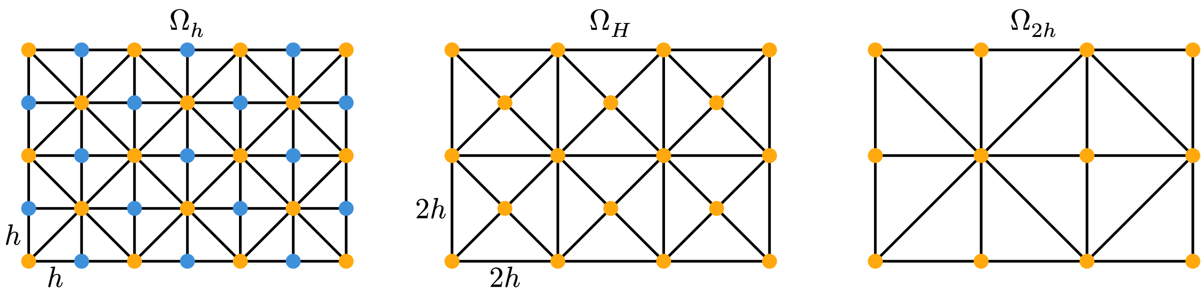


Figure 1: Grid hierarchy with the grid Ω_h on level q, Ω_H on level $q - 1$ and Ω_{2h} on level $q - 2$.

Here, the function space S_h is the finite element space corresponding to the fine grid Ω_h. It contains piecewise linear functions, with one degree of freedom at each inner grid point. Additionally, due to the Dirichlet boundary condition, they must be zero on the boundary.

Furthermore, we define both a norm and a semi-norm. The norm ("energy norm") is given by

$$\|v\| := \sqrt{a(v, v)}. \tag{6}$$

Therefore,

$$\begin{aligned}
\|v\|^2 = a(v, v) &= \int_\Omega v_\xi^2 + v_\eta^2 \\
&= \sum_\nu \int_\nu v_\xi^2 + v_\eta^2 \\
&= \sum_\nu \left(v_\xi^2\big|_\nu + v_\eta^2\big|_\nu \right) \underbrace{\frac{1}{2} h^2}_{\text{area}(\nu)} \\
&= \sum_\nu \left(\frac{(v_{\text{east}} - v_{\text{center}})^2}{h^2} + \frac{(v_{\text{north}} - v_{\text{center}})^2}{h^2} \right) \frac{1}{2} h^2 \\
&= \sum_\nu \frac{1}{2} (v_{\text{east}} - v_{\text{west}})^2 + \sum_\nu \frac{1}{2} (v_{\text{north}} - v_{\text{south}})^2 \\
&= \sum_{\substack{i,j \in \Omega_h \\ d(i,j) = h}} (v_i - v_j)^2.
\end{aligned} \tag{7}$$

The last step is justified because each vertical and horizontal border is adjacent to two different triangles (accounting for the factor 2). Note that each pair (i, j) is only used once, meaning that if we include pair (i, j) in the sum, we do not include (j, i) as well.

Similarly, we define a seminorm (i.e., something that looks like a norm but that could evaluate to zero, $|x| = 0$, even though $x \neq 0$),

$$|v|^2 := \sum_{\substack{i,j \in \Omega_H \\ d(i,j) = 2h}} \frac{1}{2} (v_i - v_j)^2. \tag{8}$$

Note that for both norms the sum is only taken over all vertical and horizontal edges of the triangulation (i.e., pairs (i, j) with Euclidean distance h), not over the diagonal edges.

For any $p_i, p_j \in \Omega_H$ with $d(i, j) = 2h$, the midpoint p_k between them lies in Ω_h and satisfies $d(i, k) = d(k, j) = h$. Since

$$(u_i - u_j)^2 = ((u_i - u_k) + (u_k - u_j))^2 \leq 2(u_i - u_k)^2 + 2(u_k - u_j)^2 \tag{9}$$

(this follows from $((u_i - u_k) - (u_k - u_j))^2 \geq 0$), summing over all such pairs gives

$$|u|^2 = \sum_{\substack{i,j \in \Omega_H \\ d(i,j) = 2h}} \frac{1}{2} (u_i - u_j)^2 \leq \sum_{\substack{i,j \in \Omega_H \\ d(i,j) = 2h}} (u_i - u_k)^2 + (u_k - u_j)^2. \tag{10}$$

Every point $p_k \in \Omega_h \setminus \Omega_H$ has exactly two $2h$-edges of Ω_H passing through it, contributing exactly its four incident h-edges of Ω_h. Hence, summing over all pairs $(i, j) \in \Omega_H$, every h-edge (i, k) with $p_k \in \Omega_h \setminus \Omega_H$ appears exactly once on the right-hand side, so it equals $\|u\|^2$. Therefore,

$$|u| \leq \|u\| \qquad \forall u \in S_h. \tag{11}$$

2. Decomposition

$$\begin{aligned} S_h &: \text{finite element space on } \Omega_h \\ S_H &: \text{finite element space on } \Omega_H \\ T_h &:= \{ w \in S_h \mid w(p) = 0 \qquad \forall p \in \Omega_H \} \end{aligned} \tag{12}$$

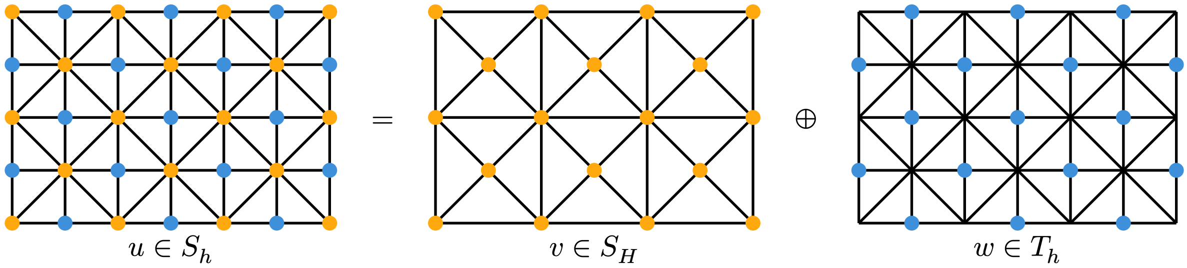


Figure 2: The space S_h of the fine grid (left) can be expressed as a sum of the space S_H on the coarse grid (middle) and the space T_h (right).

$$S_h = S_H \oplus T_h \qquad u = v + w \tag{13}$$

The basis functions of T_h form pyramid-like patterns. As a result, all functions in T_h are zero on the diagonal lines through the coarse grid points.

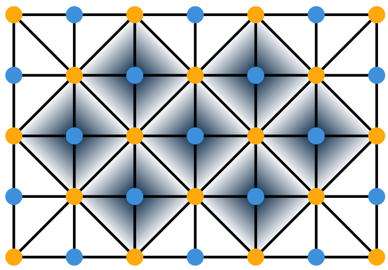
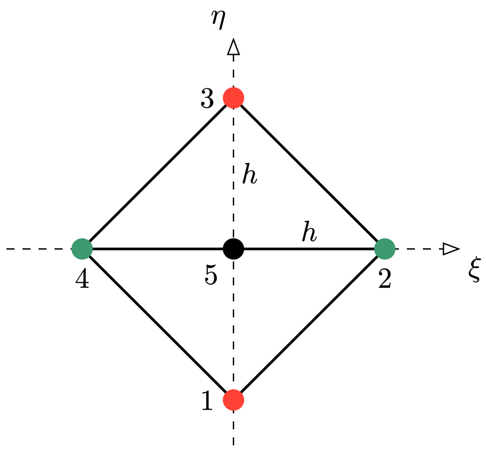


Figure 3: Each shaded area corresponds to one of the basis functions of T_h. The supports of the respective basis functions are disjoint (they do not overlap each other).

3. Bounding the Bilinear Form $a(v, w)$



Here, $\mathrm{I} = \triangle\, 124$ and $\mathrm{II} = \triangle\, 423$.

$$a(v, w)_{\mathrm{I}} = \int_{\mathrm{I}} v_\xi w_\xi + v_\eta w_\eta = \int_{\mathrm{I}} v_\xi w_\xi + \int_{\mathrm{I}} v_\eta w_\eta \tag{14}$$

$$\int_{\mathrm{I}} v_\xi w_\xi = \int_{I_{\text{left}}} v_\xi w_\xi + \int_{I_{\text{right}}} v_\xi w_\xi = 0 \tag{15}$$

da $w_\xi\big|_{I_{\text{left}}} = -w_\xi\big|_{I_{\text{right}}}$ und $v_\xi = \text{const.}$

$$\begin{aligned}
\int_{\mathrm{I}} v_\eta w_\eta &= \int_{\mathrm{I}} \frac{1}{2}v_\eta^2 - \frac{1}{2}v_\eta^2 + \frac{1}{2}w_\eta^2 - \frac{1}{2}w_\eta^2 + v_\eta w_\eta \\
&= \int_{\mathrm{I}} \frac{1}{2}\left(-v_\eta^2 + 2v_\eta w_\eta - w_\eta^2\right) + \frac{1}{2}v_\eta^2 + \frac{1}{2}w_\eta^2 \\
&= \int_{\mathrm{I}} \frac{1}{2}v_\eta^2 + \frac{1}{2}w_\eta^2 - \frac{1}{2}\left(v_\eta^2 - 2v_\eta w_\eta + w_\eta^2\right) \\
&= \frac{1}{2}\int_{\mathrm{I}} v_\eta^2 + \frac{1}{2}\int_{\mathrm{I}} w_\eta^2 - \frac{1}{2}\int_{\mathrm{I}} \left(v_\eta + w_\eta\right)^2 \qquad (16) \\
&= \frac{1}{2}\int_{\mathrm{I}} v_\eta^2 + \frac{1}{2}\int_{\mathrm{I}} v_\xi^2 - \frac{1}{2}\int_{\mathrm{I}} v_\xi^2 + \frac{1}{4}\int_{\mathrm{I}}\left(w_\eta^2 + w_\eta^2\right) - \frac{1}{2}\int_{\mathrm{I}} \left(v_\eta + w_\eta\right)^2 \\
&= \underbrace{\frac{1}{2}\int_{\mathrm{I}}\left(v_\eta^2 + v_\xi^2\right)}_{A_{\mathrm{I}}} + \underbrace{\frac{1}{4}\int_{\mathrm{I}}\left(w_\eta^2 + w_\xi^2\right)}_{B_{\mathrm{I}}} - \underbrace{\frac{1}{2}\int_{\mathrm{I}} v_\xi^2}_{C_{\mathrm{I}}} - \underbrace{\frac{1}{2}\int_{\mathrm{I}} \left(v_\eta + w_\eta\right)^2}_{D_{\mathrm{I}}} \\
&= A_{\mathrm{I}} + B_{\mathrm{I}} - C_{\mathrm{I}} - D_{\mathrm{I}}
\end{aligned}$$

When summing up the term A for all triangles in the triangulation, we simply get

$$\frac{1}{2}\int_\Omega v_\eta^2 + v_\xi^2 = \frac{1}{2}\, a(v,v) = \frac{1}{2}\,\|v\|^2. \tag{17}$$

Similarly, for B:

$$\frac{1}{4}\int_\Omega w_\eta^2 + w_\xi^2 = \frac{1}{4}\, a(w,w) = \frac{1}{4}\,\|w\|^2. \tag{18}$$

Since v is linear in both $\mathrm{I} = \triangle\, 124$ and in $\mathrm{II} = \triangle\, 423$, v_ξ is a constant in both triangles. However, because of the shared boundary $\overline{42}$, we know that $v_\xi\big|_{\mathrm{I}} = v_\xi\big|_{\mathrm{II}} = \frac{v_2 - v_4}{2h}$.

For C we find

$$\sum_\nu C_\nu = \sum_{\substack{i,j \in \Omega_{2h} \\ d(i,j) = 2h}} \frac{1}{4}\left(v_i - v_j\right)^2 \tag{19}$$

We can also show that

$$\sum_\nu D_\nu \geq \sum_{\substack{i,j \in \Omega_{2h} \setminus \Omega_H \\ d(i,j) = 2h}} \frac{1}{4}\left(v_i - v_j\right)^2 \tag{20}$$

Together with C, we get

$$
\begin{aligned}
\sum_\nu C_\nu + D_\nu &\geq \sum_{\substack{i,j\in\Omega_{2h}\\ d(i,j)=2h}} \frac{1}{4}\left(v_i - v_j\right)^2 + \sum_{\substack{i,j \text{ red}\\ d(i,j)=2h}} \frac{1}{4}\left(v_i - v_j\right)^2 \\
&= \sum_{\substack{i,j\in\Omega_H\\ d(i,j)=2h}} \frac{1}{4}\left(v_i - v_j\right)^2 \\
&= \frac{1}{2} \sum_{\substack{i,j\in\Omega_H\\ d(i,j)=2h}} \frac{1}{2}\left(v_i - v_j\right)^2 \\
&= \frac{1}{2}\ |v|^2
\end{aligned}
\tag{21}
$$

according to the definition of the seminorm $|\cdot|$.

When summing up the terms for A (Equation 17), B (Equation 18), C and D (Equation 21), we find that

$$
a(v,w) \leq \underbrace{\frac{1}{2}\ \|v\|^2}_{A} + \underbrace{\frac{1}{4}\ \|w\|^2}_{B} - \underbrace{\frac{1}{2}\ |v|^2}_{C+D} = \frac{1}{2}\left(\|v\|^2 - |v|^2\right) + \frac{1}{4}\ \|w\|^2. \tag{22}
$$

Notice that we have an inequality ($\leq$) here because of the bound on $\sum_\nu D_\nu$.

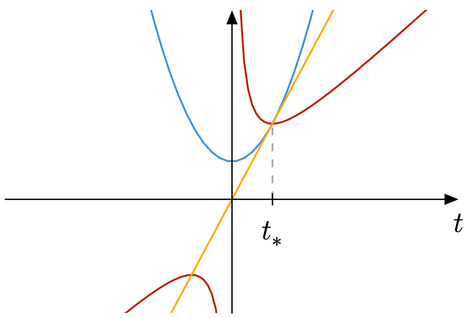
Interestingly, this upper bound is quadratic in $\|w\|$ but at the same time, bilinear form $a(v,w)$ is linear in w since $a(v,tw) = t\,a(v,w)$ for $t \in \mathbb{R}$ (homogeneity). If we set $w =: t\hat{w}$ such that $\|\hat{w}\| = 1$ and $t > 0$, we get

$$
\begin{aligned}
a(v,t\hat{w}) = t\,a(v,\hat{w}) &\leq \frac{1}{2}\left(\|v\|^2 - |v|^2\right) + \frac{1}{4}t^2 \qquad \mid : t \\
a(v,\hat{w}) &\leq \frac{\|v\|^2 - |v|^2}{2t} + \frac{1}{4}t.
\end{aligned}
\tag{23}
$$

We substitute $\mu := \|v\|^2 - |v|^2$ for brevity. This function has the following minimum

$$
\begin{aligned}
\frac{\partial}{\partial t}\left(\frac{\mu}{2t} + \frac{1}{4}t\right) &= -\frac{\mu}{2t^2} + \frac{1}{4} \overset{!}{=} 0 \\
t_* &= \sqrt{2\mu}.
\end{aligned}
\tag{24}
$$

This is the value where the tangent $f(t) = Ct$ (yellow) touches the quadratic function $\frac{1}{2}\mu + \frac{1}{4}t^2$ (blue). The derivative is shown in red.



Accordingly,

$$
\begin{aligned}
a(v,\hat{w}) &\leq \frac{\mu}{2t_*} + \frac{1}{4}t_* \\
&= \frac{\mu}{2\sqrt{2\mu}} + \frac{1}{4}\sqrt{2\mu} \\
&= \frac{1}{\sqrt{8}}\sqrt{\mu} + \frac{\sqrt{2}}{\sqrt{16}}\sqrt{\mu} \\
&= \sqrt{\frac{1}{2}\mu}.
\end{aligned} \tag{25}
$$

Now we multiply on both sides by $t = \|w\| > 0$ and we get

$$
t\, a(v,\hat{w}) = a(v, t\,\hat{w}) = a(v,w) \leq \sqrt{\frac{1}{2}\mu}\ \|w\|. \tag{26}
$$

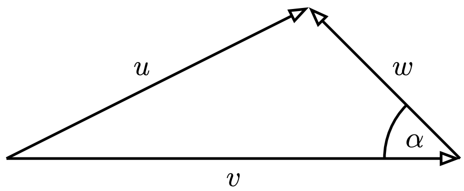
Because of homogeneity, when we negate w, we get

$$
-a(v,w) = a(v,-w) \leq \sqrt{\frac{1}{2}\mu}\ \|-w\| = \sqrt{\frac{1}{2}\mu}\ \|w\|. \tag{27}
$$

Therefore, we can replace $a(v,w)$ by its absolute value. Finally, this yields

$$
|a(v,w)| \leq \sqrt{\frac{1}{2}\left(\|v\|^2 - |v|^2\right)}\ \|w\| = \sqrt{\frac{1}{2}\left(1 - \frac{|v|^2}{\|v\|^2}\right)}\ \|v\|\|w\|. \tag{28}
$$

Geometric Interpretation



We decompose u into the sum $v + w$. These vectors are part of the same Hilbert space with inner product $a(\cdot,\cdot)$. We define the angle α as

$$
\cos(\alpha) = \frac{a(v,w)}{\|v\|\|w\|}. \tag{29}
$$

Therefore, lemma 3.1 can be written in the following way,

$$
|\cos(\alpha)| = \frac{|a(v,w)|}{\|v\|\|w\|} \leq \sqrt{\frac{1}{2}\left(1 - \left(\frac{|v|}{\|v\|}\right)^2\right)} \tag{30}
$$

or, alternatively,

$$
\cos^2(\alpha) \leq \frac{1}{2}(1 - \lambda^2) \tag{31}
$$

where $\lambda := \frac{|v|}{\|v\|}$. Due to the identity $\sin^2\alpha + \cos^2\alpha = 1$, we can rearrange this inequality,

$$
\cos^2\alpha = 1 - \sin^2\alpha \leq \frac{1}{2}(1-\lambda^2) \qquad \Longrightarrow \qquad \sin^2\alpha \geq \frac{1}{2}(1+\lambda^2). \tag{32}
$$

Furthermore, we observe that for a fixed v and a fixed α, u is the shortest when $u \perp w$. In this case, $\sin(\alpha) = \frac{\|u\|}{\|v\|}$. Generally, $\|u\| \geq \|v\| \sin(\alpha)$. Since u and v have the same values on the coarse points, we know that $|u| = |v|$.

$$|u| = |v| = \lambda\|v\| \leq \frac{\lambda}{\sin(\alpha)}\|u\| \leq \frac{\lambda}{\sqrt{\frac{1}{2}(1+\lambda^2)}}\|u\| \tag{33}$$

Finally, we express this as

$$\frac{|u|}{\|u\|} \leq \frac{\lambda}{\sqrt{\frac{1}{2}(1+\lambda^2)}} = \sqrt{\frac{\lambda^2}{\frac{1}{2}(1+\lambda^2)}} = \sqrt{\frac{2\lambda^2}{1+\lambda^2}} =: \sqrt{\rho} \tag{34}$$

4. Gauss-Seidel Relaxation

We use a finite element discretization. Interestingly, the stencil is the same for all inner grid points, whether in Ω_h or in Ω_H:

$$\begin{pmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{pmatrix} \tag{35}$$

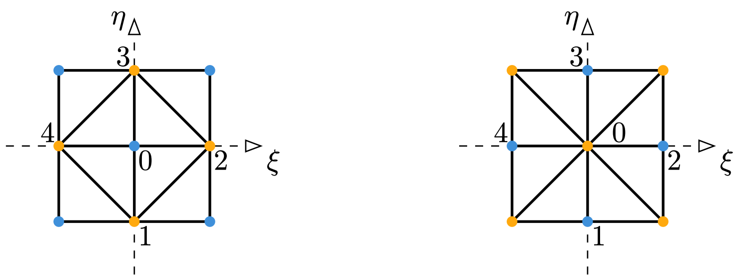


Figure 7: Neighborhood of a fine grid node in $\Omega_h \setminus \Omega_H$ (left) and of a coarse node Ω_H (right).

The stencil results in the following Gauss-Seidel relaxation for a single row:

$$u_i = \frac{1}{4}\sum_{j=1}^{4} u_j + b_i \tag{36}$$

where $b_i = \int_\Omega f\varphi_i$ is the right-hand side. In the following section, we will be concerned with the error of our approximation for u. Therefore, we assume that $b = 0$.

We split the Gauss-Seidel relaxation into two steps ($\triangleq$ red-black Gauss-Seidel). First, we will smooth the fine points ($\in \Omega_h \setminus \Omega_H$), then the coarse points ($\in \Omega_H$). Formally,

$$\begin{aligned} \left(G_h^{\mathrm{I}} u\right)_i &= \begin{cases} u_i & p_i \in \Omega_H & \text{(yellow)} \\ \frac{1}{4}\sum_{j\in N(i)} u_j & p_i \in \Omega_h \setminus \Omega_H & \text{(blue)} \end{cases} \\ \left(G_h^{\mathrm{II}} u\right)_i &= \begin{cases} \frac{1}{4}\sum_{j\in N(i)} u_j & p_i \in \Omega_H & \text{(yellow)} \\ u_i & p_i \in \Omega_h \setminus \Omega_H & \text{(blue)} \end{cases} \end{aligned} \tag{37}$$

Here, $N(i)$ is the set of neighboring points with distance h to p_i. The operator $G_h^{\mathrm{I}} \cdot G_h^{\mathrm{II}}$ represents a full Gauss-Seidel step.

Lemma 4.1

Assume that Ω is convex. We want to show: if $u = G_h^{\mathrm{I}} u$ then

$$\|G_h^{\mathrm{II}} u\| \leq |u|. \tag{38}$$

half a Gauss-Seidel step decreases the energy norm beyond the seminorm - this means it is reduced by at least the factor $\sqrt{\rho}$.

Proof. For convenience we first ignore the boundary $\partial\Omega$; we comment on the necessary modifications at the end of the proof.

Let $\bar{u} := G_h^{\mathrm{II}} u$, and fix a point $p_0 \in \Omega_H$ with its four neighbors $1, 2, 3, 4$ at distance h, which necessarily lie in $\Omega_h \setminus \Omega_H$ (cf. Figure 7, right).

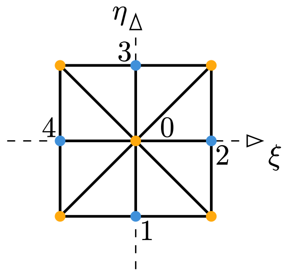


Figure 8: Point $p_0 \in \Omega_H$ and its four neighbors $1, 2, 3, 4 \in \Omega_h \setminus \Omega_H$ at distance h

For arbitrary real numbers z_1, z_2, z_3, z_4 one has the algebraic identity

$$2 \sum_{j=1}^{4} z_j^2 = \\ (z_1 - z_2)^2 + (z_2 - z_3)^2 + (z_3 - z_4)^2 + (z_4 - z_1)^2 + \frac{1}{2}(z_1 + z_2 + z_3 + z_4)^2 - \frac{1}{2}(z_1 - z_2 + z_3 - z_4)^2. \tag{39}$$

Since $\bar{u} = G_h^{\mathrm{II}} u$, the value at p_0 is by definition the mean of the (unaffected) values at its neighbors,

$$\bar{u}_0 = \frac{1}{4} \sum_{i=1}^{4} \bar{u}_i = \frac{1}{4} \sum_{i=1}^{4} u_i, \tag{40}$$

because G_h^{II} leaves the values at $\Omega_h \setminus \Omega_H$ unchanged. Put $z_i := \bar{u}_i - \bar{u}_0 = u_i - \bar{u}_0$. Then

$$\sum_{j=1}^{4} z_j = \sum_{j=1}^{4} u_j - 4\bar{u}_0 = 0, \tag{41}$$

so the first-order term $\frac{1}{2}(z_1 + z_2 + z_3 + z_4)^2$ in Equation 39 vanishes, and dropping the remaining non-positive term $-\frac{1}{2}(z_1 - z_2 + z_3 - z_4)^2$ only weakens the identity into an inequality. We obtain

$$\sum_{j=1}^{4} \left(\bar{u}_j - \bar{u}_0\right)^2 \leq \frac{1}{2}\left[(u_1 - u_2)^2 + (u_2 - u_3)^2 + (u_3 - u_4)^2 + (u_4 - u_1)^2\right]. \tag{42}$$

Recall from the definition of the energy norm (2.3) that, splitting the sum according to which grid the points belong to,

$$\|\bar{u}\|^2 = \sum_{i \in \Omega_H} \sum_{\substack{j \in \Omega_h \setminus \Omega_H \\ d(i,j) = h}} \left(\bar{u}_i - \bar{u}_j\right)^2. \tag{43}$$

For fixed $i = p_0$, the inner sum is exactly the left-hand side of Equation 42. Summing the bound Equation 42 over all $p_0 \in \Omega_H$ therefore gives

$$\|G_h^{\mathrm{II}} u\|^2 \le \sum_{\substack{k,l \in \Omega_h \setminus \Omega_H \\ d(k,l)=H}} (u_k - u_l)^2, \tag{44}$$

where $H = \sqrt{2}h$ is the distance between two opposite corners of the diamond in Figure 8.

Every pair $(k, l) \in \Omega_h \setminus \Omega_H$ with $d(k, l) = H$ occurs in the diamonds of **two** neighboring points $p_0 \in \Omega_H$. The factor $\frac{1}{2}$ on the right-hand side of Equation 42 exactly compensates for this double counting, so no extra factor appears in Equation 44.

This is the only place where the hypothesis $u = G_h^{\mathrm{I}} u$ enters. Since $u = G_h^{\mathrm{I}} u$, every point $k \in \Omega_h \setminus \Omega_H$ already satisfies the relaxation equation

$$u_k = \frac{1}{4} \sum_{j \in N(k)} u_j, \tag{45}$$

i.e. u_k is the mean of its four neighbors, which lie in Ω_H.

Consider a pair $k, l \in \Omega_h \setminus \Omega_H$ with $d(k, l) = H$, and let $1, 3, 5, 7 \in \Omega_H$ be enumerated so that $1, 5$ are the neighbors shared in the relevant direction (cf. Figure 4 of Braess' paper). Using the mean-value property at k and l,

$$16(u_k - u_l)^2 = (u_1 + u_5 - u_3 - u_7)^2 \le 2(u_1 - u_3)^2 + 2(u_5 - u_7)^2, \tag{46}$$

where the last step is the elementary inequality $(a + b)^2 \le 2a^2 + 2b^2$ applied to $a = u_1 - u_3$, $b = u_5 - u_7$.

Summing this over all pairs (k, l) from Equation 44 — each pair $(u_1 - u_3)^2$ now reappears for two different pairs (k, l), which compensates the factor $\frac{1}{4}$ — we obtain

$$\|G_h^{\mathrm{II}} u\|^2 \le \frac{1}{4} \sum_{\substack{m,n \in \Omega_H \\ d(m,n)=2H}} (u_m - u_n)^2. \tag{47}$$

The points m, n are again in Ω_H, but now at distance $2H = 2\sqrt{2}h$: these are exactly the two diagonals of a square of side length $2h$ with corners in Ω_H.

It remains to bound such diagonals by sides of length $2h$. Let $1, 2, 3, 4 \in \Omega_H$ be the corners of such a square, in cyclic order, so that $(u_1 - u_3)^2$ and $(u_2 - u_4)^2$ are the two diagonal terms appearing in Equation 47. A direct expansion gives the identity

$$\begin{gathered} (u_1 - u_3)^2 + (u_2 - u_4)^2 = \\ (u_1 - u_2)^2 + (u_2 - u_3)^2 + (u_3 - u_4)^2 + (u_4 - u_1)^2 - (u_1 - u_2 + u_3 - u_4)^2, \end{gathered} \tag{48}$$

and dropping the (non-positive) last term yields

$$(u_1 - u_3)^2 + (u_2 - u_4)^2 \le (u_1 - u_2)^2 + (u_2 - u_3)^2 + (u_3 - u_4)^2 + (u_4 - u_1)^2, \tag{49}$$

i.e. the sum of the two diagonals is bounded by the sum of the four sides of the square (each of length $2h$).

Every side of length $2h$ is shared by exactly two neighboring squares, so summing this inequality over all squares with corners in Ω_H and combining with Equation 47 gives

$$\|G_h^{\mathrm{II}} u\|^2 \leq \frac{1}{2} \sum_{\substack{n,m \in \Omega_H \\ d(n,m)=2h}} (u_n - u_m)^2 = |u|^2, \tag{50}$$

which is exactly the definition of the seminorm $|\cdot|$ from (2.4). This proves

$$\|G_h^{\mathrm{II}} u\| \leq |u|. \tag{51}$$

Near $\partial\Omega$, some of the neighbors $1, 2, 3, 4$ (or $1, 3, 5, 7$) used above may not exist inside Ω; by the homogeneous Dirichlet condition, the corresponding values are simply replaced by 0. The same telescoping arguments go through verbatim with these zero values inserted — only the index bookkeeping becomes more tedious, while every inequality used above (Equation 39, the mean-value property, the diagonal identity) remains valid for boundary configurations as well, consistent with how the doubled weight in (2.5) was introduced for the seminorm. ■

As a consequence of Lemma 4.1 we have the following corollary.

Corollary 4.2. If Ω is convex, then $\|G_h^{\mathrm{II}} \cdot G_h^{\mathrm{I}} u\| \leq |u|$ for every $u \in S_h$ — i.e. the hypothesis $u = G_h^{\mathrm{I}} u$ in Lemma 4.1 can be dropped, at the price of applying G_h^{I} first.

Proof. Put $u' := G_h^{\mathrm{I}} u$. Since G_h^{I} is idempotent ($G_h^{\mathrm{I}} G_h^{\mathrm{I}} = G_h^{\mathrm{I}}$), we have $u' = G_h^{\mathrm{I}} u'$, so Lemma 4.1 applies to u':

$$\|G_h^{\mathrm{II}} u'\| \leq |u'|. \tag{52}$$

Now G_h^{I} only changes the w-part of the decomposition $u = v + w$ (it leaves the values at Ω_H, i.e. the v-part, untouched), and the seminorm $|\cdot|$ depends only on the v-part by (3.3). Hence

$$|u'| = |G_h^{\mathrm{I}} u| = |u|, \tag{53}$$

and combining the two displays gives $\|G_h^{\mathrm{II}} G_h^{\mathrm{I}} u\| \leq |u|$ for every $u \in S_h$. ■

We note that the constant 1 in (4.2) is best possible, since the convergence rate of the (pure) Gauss-Seidel iteration is known to be close to 1, with $\|G_h^{\mathrm{II}} \cdot G_h^{\mathrm{I}}\| = 1 - O(h^2)$.

5. The Multigrid Algorithm

The multigrid algorithm consists of the presmoother, the coarse grid correction and the postsmoother.

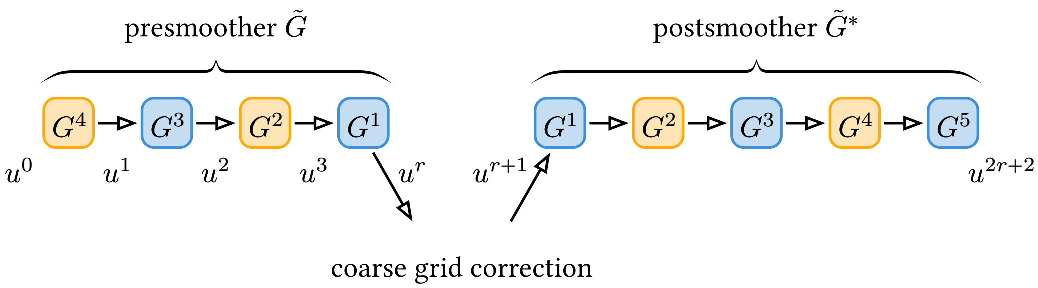


Figure 9: Schematic representation of the multigrid algorithm for $r = 4$.[1]

The individual steps in the smoothers are the Gauss-Seidel half-steps.

$$G^k := \begin{cases} G_h^{\mathrm{I}} & k \text{ odd} \\ G_h^{\mathrm{II}} & k \text{ even} \end{cases} \tag{54}$$

[1]In the first iteration, we prepend G^{r+1} to the presmoother.

In the presmoother, we have r half-steps,

$$u^r = G^r G^{r-1} \dots G^1 u^0 = \tilde{G} u^0. \tag{55}$$

The only exception is iteration 1 (i.e., the first V-cycle or W-cycle) where we prepend the half-step G^{r+1} to achieve symmetry between presmoother and postsmoother. The postsmoother contains $r + 1$ half-steps,

$$u^{2r+2} = G^1 G^2 \dots G^r G^{r+1} u^{r+1} = \tilde{G}^* u^{r+1} \tag{56}$$

In part 1 we have already shown the following three statements:

$$|a(v,w)| \le \sqrt{\frac{1}{2}\left(1 - \frac{|v|^2}{\|v\|^2}\right)}\ \|v\|\ \|w\| \qquad v \in S_H, w \in T_h, \tag{57}$$

$$|u| \le \sqrt{\rho}\ \|u\| \qquad \text{with} \qquad \rho := \frac{2\lambda^2}{\lambda^2 + 1} \text{ and } \lambda := \frac{|v^r|}{\|v^r\|}, \tag{58}$$

$$\text{if} \qquad u = G_h^{\mathrm{I}} u \qquad \text{then} \qquad \|G_h^{\mathrm{II}} u\| \le |u|. \tag{59}$$

Now, we are looking for an upper bound of the convergence rate,

$$\|u^{2r+2}\| \le \delta_q\ \|u^0\| \qquad \text{with} \qquad \delta_q < 1. \tag{60}$$

To this end, we first find the convergence rate of the presmoother (Section 6), then we include the coarse grid correction (Section 7 and Section 8), and finally look at a full multigrid iteration (Section 9).

6. Presmoother

From Equation 58 and Equation 59, we find that if $u = G_h^{\mathrm{I}} u$, then

$$\|G_h^{\mathrm{II}} u\| \le |u| \le \sqrt{\rho}\ \|u\|. \tag{61}$$

Unfortunately, this only gives us an upper bound for the convergence rate of G_h^{II} and not for G_h^{I}. Therefore, we will prove the following two statements

$$\begin{aligned} \text{if } u = G_h^{\mathrm{II}} u: &\qquad \frac{\|G_h^{\mathrm{I}} u\|}{\|u\|} \le \frac{\|G_h^{\mathrm{II}} G_h^{\mathrm{I}} u\|}{\|G_h^{\mathrm{I}} u\|} \\ \text{if } u = G_h^{\mathrm{I}} u: &\qquad \frac{\|G_h^{\mathrm{II}} u\|}{\|u\|} \le \frac{\|G_h^{\mathrm{I}} G_h^{\mathrm{II}} u\|}{\|G_h^{\mathrm{II}} u\|} \end{aligned} \tag{62}$$

First, we examine the G_h^{I} in more detail.

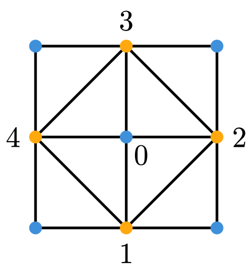


Figure 10: Neighborhood of a blue point with grid point numbering.

In this neighborhood, G_h^{I} essentially computes the average of the four neighbors,

$$u_0 \leftarrow \frac{1}{4}(u_1 + u_2 + u_3 + u_4). \tag{63}$$

This operation minimizes the energy function $J(v)$ over the set $u + T_h$, since

$$J(v) = a(v,v) + \cancel{(f,v)_0} = \|v\|^2 = \sum_{\substack{i,j \\ d(i,j)=h}} \left(v_i - v_j\right)^2. \tag{64}$$

The part of the sum relevant for the neighborhood in Figure 10 is

$$\left(v_1 - v_0\right)^2 + \left(v_2 - v_0\right)^2 + \left(v_3 - v_0\right)^2 + \left(v_4 - v_0\right)^2. \tag{65}$$

This term can be interpreted as a function of v_0 (as G_h^{I} only changes the values at the blue points). The minimum of this function can be computed by evaluating its derivative by v_0.

$$\begin{aligned} 2(v_0 - v_1) + 2(v_0 - v_2) + 2(v_0 - v_3) + 2(v_0 - v_4) = 8v_0 - 2\sum_{j=1}^{4} v_j \overset{!}{=} 0 \\ \Longrightarrow v_0 = \frac{1}{4}(v_1 + v_2 + v_3 + v_4) \end{aligned} \tag{66}$$

This is exactly what G_h^{I} computes at each blue grid point, minimizing the sum in Equation 64.

Now we define $v \coloneqq G_h^{\mathrm{I}} u$. Since v is the minimizer of $J(u + w) \quad \forall w \in T_h$, we can state that

$$J(v + w) \geq J(v) \qquad \forall w \in T_h, \tag{67}$$

or alternatively,

$$\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(v + \varepsilon w)\right|_{\varepsilon=0} = 0 \qquad \forall w \in T_h, \varepsilon \in \mathbb{R}. \tag{68}$$

Using the bilinear form $a(\cdot,\cdot)$, we expand this term as follows:

$$\begin{aligned} J(v + \varepsilon w) &= a(v + \varepsilon w, v + \varepsilon w) \\ &= a(v,v) + 2\varepsilon a(v,w) + \varepsilon^2 a(w,w). \end{aligned} \tag{69}$$

Then, the derivative is

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(v + \varepsilon w) = 2a(v,w) + 2\varepsilon a(w,w). \tag{70}$$

For $\varepsilon = 0$, we get

$$\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(v + \varepsilon w)\right|_{\varepsilon=0} = 2a(v,w) \overset{!}{=} 0 \qquad \forall w \in T_h. \tag{71}$$

Since $a(\cdot,\cdot)$ has the role of the inner product, we find that v is perpendicular to T_h. If $T_h^{\perp} \coloneqq \{u \in S_h \mid (u,w) = 0 \quad \forall w \in T_h\}$ is the orthogonal complement of T_h, then $v \in T_h^{\perp}$.

Let P_{T_h} and $P_{T_h^{\perp}}$ denote the orthogonal projections onto T_h and $T_h^{\perp}$, respectively. G_h^{I} only changes degrees of freedom that are part of T_h. Thus,

$$P_{T_h^{\perp}} u = P_{T_h^{\perp}} v = v = G_h^{\mathrm{I}} u. \tag{72}$$

It follows that G_h^{I} is the orthogonal projection onto $T_h^{\perp}$.

This idea can also be visualized using a two-dimensional S_h and one-dimensional T_h and $T_h^{\perp}$.

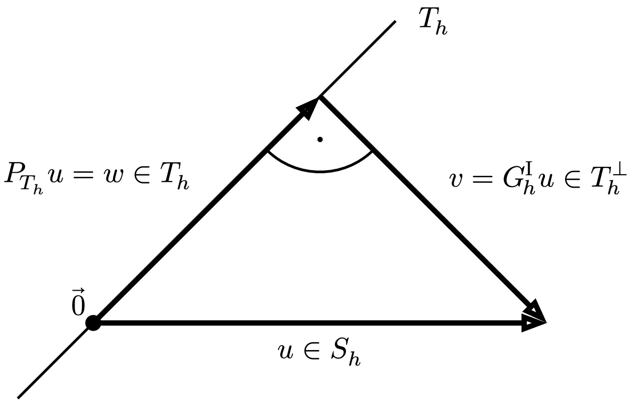


Figure 11: Visualization of the operator $G_h^{\mathrm{I}} = P_{T_h^\perp}$.

Since $T_h \oplus T_h^\perp = S_h$, we can find a $w \in T_h$ such that $u - w = v$. Since $v \perp w \ \forall w \in T_h$, it follows that $w = P_{T_h} u$.

G_h^{I} can also be expressed in terms of P_{T_h}:

$$v = u - w = u - P_{T_h} u = \left(I - P_{T_h}\right) u = G_h^{\mathrm{I}} u, \tag{73}$$

where I is the identity operator.

The projection theorem tells us that the split of u into $u = w + v$ is unique.

Analogously, it can be shown that G_h^{II} is an orthogonal projection. In this case, instead of T_h we need to use the space $\hat{T}_h := \{w \in S_h \mid w(p_i) = 0 \text{ if } p_i \text{ blue}\}$

A fundamental property of orthogonal projections is that they are self-adjoint.

An operator Q^* is the adjoint of Q if

$$(Qu, v) = (u, Q^* v) \qquad \forall u, v. \tag{74}$$

It is self-adjoint if $Q^* = Q$, i.e.,

$$(Qu, v) = (u, Qv) \qquad \forall u, v. \tag{75}$$

This property of orthogonal projections can be shown as follows. Let $x = x_1 + x_2 \in S_h$ and $y = y_1 + y_2 \in S_h$ such that $x_1, y_1 \in T_h$ and $x_2, y_2 \in T_h^\perp$. If P is the orthogonal projection onto T_h, then

$$\begin{aligned} (Px, y) &= (Px_1 + \cancel{Px_2}, y) = (x_1, y) = (x_1, y_1 + y_2) \\ &= (x_1, y_1) + \cancel{(x_1, y_2)} = (x_1, y_1) = (x_1, y_1) + \cancel{(x_2, y_1)} \\ &= (x_1 + x_2, y_1) = (x, y_1) = (x, Py_1 + \cancel{Py_2}) = (x, Py). \end{aligned} \tag{76}$$

The inner products (x_1, y_2) and (x_2, y_1) are zero because one of the arguments is in T_h and the other in $T_h^\perp$ (so they are orthogonal to each other).

Another property of orthogonal projections is that they are idempotent:

$$P^2 u = PPu = Pu \qquad \forall u. \tag{77}$$

Now we can prove Equation 62. Assuming $u = G_h^{\mathrm{II}} u$, we get

$$
\begin{aligned}
\|G_h^{\mathrm{I}} u\|^2 &= \left(G_h^{\mathrm{I}} u, G_h^{\mathrm{I}} u\right) && \text{definition of the norm} \\
&= \left(u, G_h^{\mathrm{I}} G_h^{\mathrm{I}} u\right) && G_h^{\mathrm{I}} \text{ self-adjoint} \\
&= \left(u, G_h^{\mathrm{I}} u\right) && G_h^{\mathrm{I}} \text{ idempotent} \\
&= \left(G_h^{\mathrm{II}} u, G_h^{\mathrm{I}} u\right) && \text{by assumption} \\
&= \left(u, G_h^{\mathrm{II}} G_h^{\mathrm{I}} u\right) && G_h^{\mathrm{II}} \text{ self-adjoint} \\
&\leq \|u\| \cdot \|G_h^{\mathrm{II}} G_h^{\mathrm{I}} u\| && \text{Cauchy-Schwarz inequality.}
\end{aligned}
\tag{78}
$$

Cauchy-Schwarz inequality:

$$
|(u, v)| \leq \|u\| \cdot \|v\| \qquad \forall u, v. \tag{79}
$$

The case of equality happens when u and v are linearly dependent,

$$
|(u, v)| = \|u\| \cdot \|v\| \qquad \Longleftrightarrow \qquad u = \alpha v \qquad \alpha \in \mathbb{R}. \tag{80}
$$

Rearranging Equation 78, we get

$$
\text{if } u = G_h^{\mathrm{II}} : \qquad \frac{\|G_h^{\mathrm{I}} u\|}{\|u\|} \leq \frac{\|G_h^{\mathrm{II}} G_h^{\mathrm{I}} u\|}{\|G_h^{\mathrm{I}} u\|}. \tag{81}
$$

Similarly, this can be shown the other way around (when we swap G_h^{I} and G_h^{II}):

$$
\text{if } u = G_h^{\mathrm{I}} : \qquad \frac{\|G_h^{\mathrm{II}} u\|}{\|u\|} \leq \frac{\|G_h^{\mathrm{I}} G_h^{\mathrm{II}} u\|}{\|G_h^{\mathrm{II}} u\|}. \tag{82}
$$

These two equations can be applied to the presmoother, which is an alternating sequence of G_h^{I} and G_h^{II}:

$$
\frac{\|u^{k+1}\|}{\|u^k\|} \leq \frac{\|u^{k+2}\|}{\|u^{k+1}\|} \qquad k = 0, ..., r-2. \tag{83}
$$

This means that the Gauss-Seidel half-steps become more ineffective over time (or in case of equality, the norm reduction stays the same). Note that the last step of the presmoother is always G_h^{I}, i.e., $u^r = G_h^{\mathrm{I}} u^{r-1}$. If we were to append another Gauss-Seidel half-step, we would have $u^{r+1} = G_h^{\mathrm{II}} u^r$. We know from Equation 61 that this step would have a norm reduction of $\sqrt{\rho}$ or less (the lower the better). Due to Equation 83, his hypothetical last step $u^{r+1} = G_h^{\mathrm{II}} u^r$ is the most "ineffective" step:

$$
\frac{\|u^{r+1}\|}{\|u^r\|} \geq \frac{\|u^{k+1}\|}{\|u^k\|} \qquad k = 1, ..., r. \tag{84}
$$

Therefore the first r half-steps also have a norm reduction of $\sqrt{\rho}$ or smaller. Consequently, after r half-steps, we have the following upper bound of the norm reduction:

$$
\|u^r\| \leq \left(\sqrt{\rho}\right)^r \, \|u^0\| = \rho^{\frac{r}{2}} \, \|u^0\|. \tag{85}
$$

7. Exact Coarse Grid Correction

First we look at an exact coarse grid correction (i.e., the equation is solved without error on the coarser grid). In this step of the multigrid algorithm, we minimize the energy function $J(v)$ over $u^r + S_H$. In other words, we evaluate

$$
u^{q-1} := \operatorname*{argmin}_{v \in S_H} J(u^r - v). \tag{86}
$$

Here, u^{q-1} is in S_H, the finite element space associated with the coarser grid on level $q-1$. In the correction step, we compute

$$u' = u^r - u^{q-1}. \tag{87}$$

Note that the exact coarse grid correction is similar to the Gauss-Seidel half steps, only that we minimize J over $u^r + S_H$ instead of $u + T_h$. Hence, using the same argument as in Equation 67 to Equation 71, we find that $u' \perp S_H$ and

$$\begin{aligned} u^{q-1} &= P_{S_H} u^r \\ u' = u^r - u^{q-1} &= I u^r - P_{S_H} u^r = \left(I - P_{S_H}\right) u^r = P_{S_H^\perp} u^r. \end{aligned} \tag{88}$$

Furthermore, as observed earlier already, the last step of the presmoother is always G_h^{I}. The resulting vector u^r is thus orthogonal to T_h (compare vector v in Figure 11).

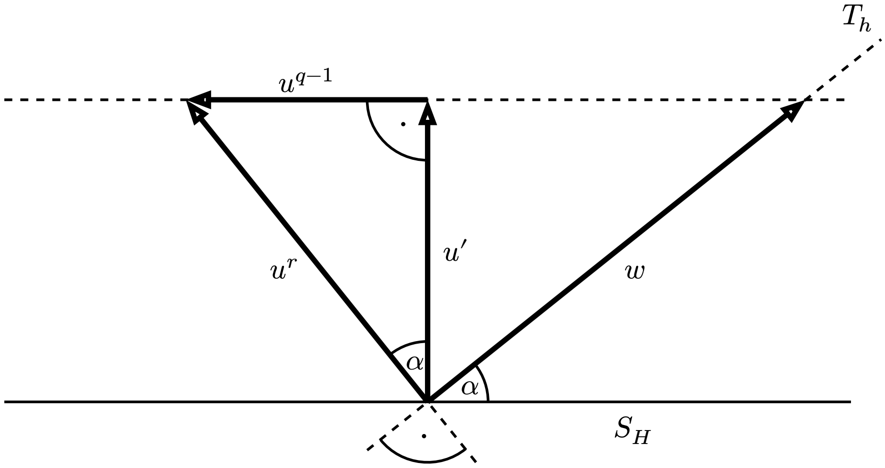


Figure 12: Visualization of the exact coarse grid correction, which is an orthogonal projection of u^r onto $S_H^\perp$, based on [2, Lemma 3.1].

From Equation 57, we find

$$\frac{(v,w)}{\|v\|\,\|w\|} = \cos(\alpha) \le \sqrt{\frac{1}{2}(1-\lambda^2)} \qquad \forall v \in S_H, w \in T_h. \tag{89}$$

Since this equation is true for all $v \in S_H$ and all $w \in T_h$, this really says something about the angle α between subspaces S_H and T_h. Furthermore, this angle is the same as the angle between u' and u^r (as a consequence of $u' \perp S_H$ and $u^r \perp T_h$).

Due to the right angle between u' and u^{q-1}, we find

$$\begin{aligned} \|u'\| = \cos(\alpha)\,\|u^r\| &\le \sqrt{\frac{1}{2}(1-\lambda^2)}\,\|u^r\| = \sqrt{\frac{\frac{1-\lambda^2}{\lambda^2+1}}{\frac{2}{\lambda^2+1}}}\,\|u^r\| \\ &= \sqrt{\frac{\frac{\lambda^2+1-2\lambda^2}{\lambda^2+1}}{\frac{2\lambda^2+2-2\lambda^2}{\lambda^2+1}}}\,\|u^r\| = \sqrt{\frac{1-\frac{2\lambda^2}{\lambda^2+1}}{2-\frac{2\lambda^2}{\lambda^2+1}}}\,\|u^r\| = \sqrt{\frac{1-\rho}{2-\rho}}\,\|u^r\|. \end{aligned} \tag{90}$$

This idea can be generalized to higher-dimensional function spaces for the following reason: Since u^{q-1} is projection of u^r onto S_H, the length of u' is the shortest distance between u^r and S_H. If we add more dimensions to S_H, the shortest distance can only become smaller but not larger. Therefore, Equation 90 still holds.

This equation gives us an upper bound for the norm reduction achieved by an exact coarse grid correction. Together with Equation 85, we find

$$\|u'\| \leq \rho^{\frac{r}{2}} \sqrt{\frac{1-\rho}{2-\rho}}\, \|u^0\| \tag{91}$$

to account for both the presmoother and the exact coarse grid correction.

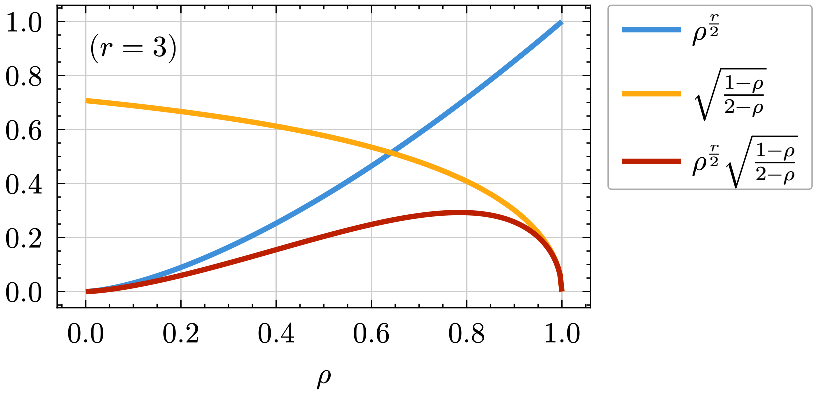


Figure 13: Norm reduction (red) after the presmoother (blue) and exact coarse grid correction (yellow) as a function of ρ.

Plotting the norm reduction of the presmoother and the exact coarse grid correction, we observe the following: In case the presmoother is not effective ($\rho^{\frac{r}{2}} = \rho = 1$), the coarse grid correction is very effective ($\sqrt{\frac{1-\rho}{2-\rho}} \to 0$). Conversely, if the coarse grid correction is rather ineffective ($\rho = 0$), the presmoother is very effective. The maximum of the red function, i.e., the worst possible norm reduction, is still far below 1.

8. Non-Exact Coarse Grid Correction

Now we generalize the coarse grid correction to include an error. Instead of evaluating u^{q-1} exactly, we compute an approximation v_1 to u^{q-1}. Assuming that we already know the convergence rate δ_{q-1} on the coarser level $q-1$, and that perform a V-cycle with one recursive multigrid call, we find that the deviation from the exact error, u^{q-1} can be bounded:

$$\|v_1 - u^{q-1}\| \leq \delta_{q-1}\, \|u^{q-1}\|. \tag{92}$$

To generalize this, we define

$$\delta := \begin{cases} 0 & q = 1 \\ \left(\delta_{q-1}\right)^{\mu} & q > 1 \end{cases} \qquad \mu := \begin{cases} 1 & \text{V-cycle} \\ 2 & \text{W-cycle} \end{cases} \tag{93}$$

and require that

$$\|v_1 - u^{q-1}\| \leq \delta\, \|u^{q-1}\|. \tag{94}$$

Remember that u^{r+1} is the actual approximation after the coarse grid correction whereas u' is the approximation if the coarse grid correction had been exact. Thus the deviation is,

$$u^{r+1} - u' = \left(u^r - v_1\right) - \left(u^r - u^{q-1}\right) = u^{q-1} - v_1. \tag{95}$$

If we define

$$v_2 := \frac{1}{\delta}\left(u^{q-1} - v_1\right) = \frac{1}{\delta}\left(u^{r+1} - u'\right) \qquad v_2 \in S_H \tag{96}$$

then (from Equation 94):

$$\begin{aligned} \|u^{r+1} - u'\| = \|u^{q-1} - v_1\| = \delta\ \|v_2\| \leq \delta\ \|u^{q-1}\| \\ \implies \|v_2\| \leq \|u^{q-1}\|. \end{aligned} \tag{97}$$

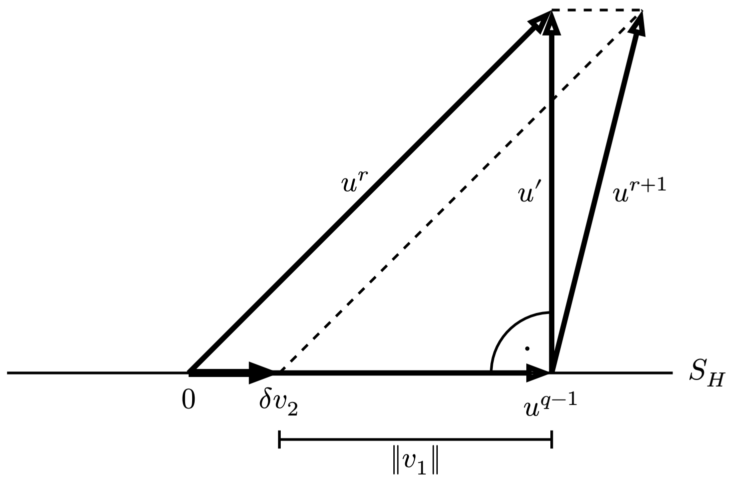


Figure 14: Visualization of the non-exact coarse grid correction.

9. Full Multigrid Iteration

To account for a full multigrid iteration, we need to include the presmoother, non-exact coarse grid correction and the postsmoother. First, let us analyze the algorithm across the first two iterations.

Iteration		1			2	
Step	presmoother	CGC	postsmoother	presmoother	CGC	postsmoother
	$G^{r+1}G^r...G^1$		$G^1...G^rG^{r+1}$	$G^r...G^1$		$G^1...G^rG^{r+1}$
Example $r = 3$	G^4 G^3 G^2 G^1		G^1 G^2 G^3 G^4	G^3 G^2 G^1		G^1 G^2 G^3 G^4

Table 1: First two iterations of the multigrid algorithm.
The coarse grid correction is abbreviated as CGC.

Note that in the first iteration, we have an additional G^{r+1} at the beginning of the presmoother. This half-step is missing in all subsequent iterations. If we were to include G^{r+1} at the beginning of the presmoother of each iteration, we would have a duplicate. For example for $r = 3$ the postsmoother of iteration 1 would end with G_h^{II} and the presmoother of iteration 2 would start with G_h^{II} again. However, since the Gauss-Seidel half-steps are idempotent (as they are projections), we have $G_h^{\mathrm{II}} \cdot G_h^{\mathrm{II}} = G_h^{\mathrm{II}}$. The additional half-step in the presmoother would not have any effect.

However, for the same reason, it is possible to include G^{r+1} conceptually at the beginning of each presmoother. Mathematically, this is the same algorithm as in Table 1. This has the benefit that now the postsmoother is the reverse of the presmoother, such that

$$\begin{aligned} \tilde{G} &= G^{r+1}G^r...G^1 && \text{presmoother} \\ \tilde{G}^* &= G^1...G^rG^{r+1} && \text{postsmoother}. \end{aligned} \tag{98}$$

As suggested by the notation, $\tilde{G}^*$ is the adjoint of $\tilde{G}$:

$$
\begin{aligned}
\left(\tilde{G}u, v\right) &= \left(G^{r+1}G^{r}...G^{1}u, v\right) \\
&= \left(G^{r}G^{r-1}...G^{1}u, G^{r+1}v\right) \\
&= \left(G^{r-1}...G^{1}u, G^{r}G^{r+1}v\right) \\
&= ... = \left(u, G^{1}G^{2}...G^{r}G^{r+1}v\right) \\
&= \left(u, \tilde{G}^{*}v\right).
\end{aligned}
\tag{99}
$$

This is true because all Gauss-Seidel half-steps G^k are self-adjoint as shown in Section 6.

Now, we are ready to take the final steps to get the convergence rate. To this end, we use the following idea: We want to get an expression for $\|u^{2r+2}\|$. This term can be expressed as follows:

$$
\max_{\hat{u}\in S_h}\left\{\frac{\left(\hat{u}, u^{2r+2}\right)}{\|\hat{u}\|}\right\} = \max_{\hat{u}\in S_h}\left\{\left(\frac{\hat{u}}{\|\hat{u}\|}, u^{2r+2}\right)\right\} = \|u^{2r+2}\|.
\tag{100}
$$

This is because (due to the Cauchy-Schwarz inequality, Equation 79)

$$
\left(\frac{\hat{u}}{\|\hat{u}\|}, u^{2r+2}\right) \leq \left\|\frac{\hat{u}}{\|\hat{u}\|}\right\| \cdot \|u^{2r+2}\| = \|u^{2r+2}\|.
\tag{101}
$$

Subsequently, we will find the upper bound

$$
\left(\frac{\hat{u}}{\|\hat{u}\|}, u^{2r+2}\right) \leq \delta_q \;\|u^0\| \qquad \forall \hat{u} \in S_h.
\tag{102}
$$

Since this upper bound must be valid for all possible $\hat{u}$, it must be valid for $\hat{u} = u^{2r+2}$ as well (when we have an equality in Equation 101). In this special case, we find

$$
\left(\frac{u^{2r+2}}{\|u^{2r+2}\|}, u^{2r+2}\right) = \|u^{2r+2}\| \leq \delta_q \;\|u^0\|.
\tag{103}
$$

To evaluate δ_q, we start with the inner product $\left(\hat{u}, u^{2r+2}\right)$.

$$
\begin{aligned}
\left(\hat{u}, u^{2r+2}\right) &= \left(\hat{u}, \tilde{G}^{*}u^{r+1}\right) && \text{Equation 56} \\
&= \left(\tilde{G}\hat{u}, u^{r+1}\right) && \text{Equation 99} \\
&= \left(\tilde{G}\hat{u}, u' + \delta v_2\right) && \text{Equation 96} \\
&= \left(\tilde{G}\hat{u}, u' - \delta u' + \delta u' + \delta v_2\right) \\
&= \left(\tilde{G}\hat{u}, (1-\delta)Qu^{r} + \delta(u' + v_2)\right).
\end{aligned}
\tag{104}
$$

Here, $Q := P^{\perp}_{S_H}$ represents the exact coarse grid correction (see Equation 88). Since it is a projection, it is idempotent ($Q^2 = Q$).

$$
\begin{aligned}
\left(\hat{u}, u^{2r+2}\right) &\leq (1-\delta)\left(\tilde{G}\hat{u}, Qu^{r}\right) + \delta\left(\tilde{G}\hat{u}, u' + v_2\right) && \text{linearity of inner product} \\
&= (1-\delta)\left(\tilde{G}\hat{u}, Q\tilde{G}u^{0}\right) + \delta\left(\tilde{G}\hat{u}, u' + v_2\right) && \text{Equation 98} \\
&\leq (1-\delta)\left(\tilde{G}\hat{u}, Q^{2}\tilde{G}u^{0}\right) + \delta \;\|\tilde{G}\hat{u}\| \;\|u' + v_2\| && \text{Equation 79.}
\end{aligned}
\tag{105}
$$

For the term $\|u' + v_2\|$ we find (see Figure 14 and Equation 97)

$$
\|u' + v_2\|^2 = \|u'\|^2 + \|v_2\|^2 \leq \|u'\|^2 + \|u^{q-1}\|^2 = \|u^{r}\|^2
\tag{106}
$$

and thus

$$\|u' + v_2\| \le \|u^r\|. \tag{107}$$

$$\begin{aligned}\left(\hat{u}, u^{2r+2}\right) &\le (1-\delta)\Big(Q\tilde{G}\hat{u}, Q\tilde{G}u^0\Big) + \delta\ \|\tilde{G}\hat{u}\|\ \|u^r\| && \text{Equation 76} \\ &\le (1-\delta)\ \|Q\tilde{G}\hat{u}\|\ \|Q\tilde{G}u^0\| + \delta\ \|\tilde{G}\hat{u}\|\ \|\tilde{G}u^0\| && \text{Equation 79.}\end{aligned} \tag{108}$$

We rearrange this term as follows. If

$$\begin{aligned} a &:= \sqrt{1-\delta}\ \|Q\tilde{G}\hat{u}\|, \qquad & c &:= \sqrt{1-\delta}\ \|Q\tilde{G}u^0\|, \\ b &:= \sqrt{\delta}\ \|\tilde{G}\hat{u}\|, \qquad & d &:= \sqrt{\delta}\ \|\tilde{G}u^0\|, \end{aligned} \tag{109}$$

then

$$\begin{aligned} (1-\delta)\ \|Q\tilde{G}\hat{u}\|\ \|Q\tilde{G}u^0\| + \delta\ \|\tilde{G}\hat{u}\|\ \|\tilde{G}u^0\| &= ac + bd = \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} c \\ d \end{pmatrix} \\ \le \left\| \begin{pmatrix} a \\ b \end{pmatrix} \right\| \cdot \left\| \begin{pmatrix} c \\ d \end{pmatrix} \right\| &= \sqrt{a^2+b^2} \cdot \sqrt{c^2+d^2} \\ = \sqrt{(1-\delta)\ \|Q\tilde{G}\hat{u}\|^2 + \delta\ \|\tilde{G}\hat{u}\|^2} &\cdot \sqrt{(1-\delta)\ \|Q\tilde{G}u^0\|^2 + \delta\ \|\tilde{G}u^0\|^2}. \end{aligned} \tag{110}$$

Let us focus on the term $(1-\delta)\ \|Q\tilde{G}u^0\|^2 + \delta\ \|\tilde{G}u^0\|^2$. From Equation 91 and Equation 85, we find

$$\begin{aligned} (1-\delta)\ \|Q\tilde{G}u^0\|^2 + \delta\ \|\tilde{G}u^0\|^2 &= (1-\delta)\ \|u'\|^2 + \delta\ \|u^r\|^2 \\ &\le (1-\delta)\left(\rho^{\frac{r}{2}}\sqrt{\frac{1-\rho}{2-\rho}}\ \|u^0\|\right)^2 + \delta\big(\rho^{\frac{r}{2}}\ \|u^0\|\big)^2 \\ &= (1-\delta)\rho^r\frac{1-\rho}{2-\rho}\ \|u^0\|^2 + \delta\rho^r\ \|u^0\|^2 \\ &= \rho^r\left((1-\delta)\frac{1-\rho}{2-\rho} + \delta\right)\ \|u^0\|^2. \end{aligned} \tag{111}$$

Similarly, for the other square root in Equation 110, we have

$$(1-\delta)\ \|Q\tilde{G}\hat{u}\|^2 + \delta\ \|\tilde{G}\hat{u}\|^2 \le \rho^r\left((1-\delta)\frac{1-\rho}{2-\rho} + \delta\right)\ \|\hat{u}\|^2. \tag{112}$$

However, ρ depends on the starting vector (previously called u^0). When we have different starting vectors, u^0 and $\hat{u}$, the corresponding ρ will be different (i.e. the value of ρ in Equation 111 may be different than in Equation 112). Furthermore, we cannot evaluate the exact value of ρ since we can start with an arbitrary start vector u^0. To overcome this problem, we assume the worst case, i.e., the maximum of the function $\rho^r\left((1-\delta)\frac{1-\rho}{2-\rho} + \delta\right)$.

To do so, we find that $0 \le \rho \le 1$. This is because

$$0 \le \lambda := \frac{|v^r|}{\|v^r\|} \le 1 \qquad \text{and} \qquad \rho := \frac{2\lambda^2}{\lambda^2+1}. \tag{113}$$

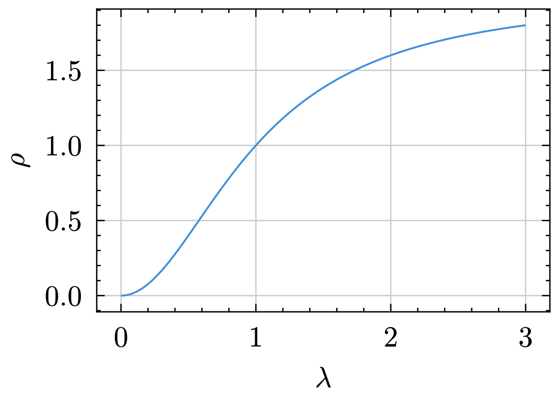


Figure 15: ρ as a function of λ.

We define

$$\delta_q := \max_{0 \le \rho \le 1} \left[\rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right) \right]. \tag{114}$$

Then

$$\begin{aligned} (1-\delta)\ \|Q\tilde{G}u^0\|^2 + \delta\ \|\tilde{G}u^0\|^2 &\le \delta_q\ \|u^0\|^2 \\ (1-\delta)\ \|Q\tilde{G}\hat{u}\|^2 + \delta\ \|\tilde{G}\hat{u}\|^2 &\le \delta_q\ \|\hat{u}\|^2. \end{aligned} \tag{115}$$

We substitute this inequality back into Equation 110:

$$\begin{aligned} (\hat{u}, u^{2r+2}) &\le \sqrt{\delta_q\ \|\hat{u}\|^2} \cdot \sqrt{\delta_q\ \|u^0\|^2} \\ &= \delta_q\ \|\hat{u}\|\ \|u^0\| \qquad \forall \hat{u} \in S_h. \end{aligned} \tag{116}$$

Using the arguments from Equation 100 to Equation 103, we find that

$$\|u^{2r+2}\| \le \delta_q\ \|u^0\| \qquad \text{with} \qquad \delta_q := \max_{0 \le \rho \le 1} \left[\rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right) \right]. \tag{117}$$

δ_q is an upper bound for the norm reduction of a single multigrid iteration on level q. This equation can now be applied for both a two-grid method and a general multigrid method to get concrete numerical values for δ_q.

10. The Two-Grid Method

In the two-grid method, we have only two different levels. During the coarse grid correction, we solve for the error exactly. Therefore (also see Equation 93),

$$0 = \|v_1 - u^{q-1}\| < \delta\ \|u^{q-1}\| \qquad \Longrightarrow \qquad \delta = 0. \tag{118}$$

In this case,

$$\delta_{\mathrm{TG}} := \delta_q = \max_{0 \le \rho \le 1} \left[\rho^r \left((1-\delta) \frac{1-\rho}{2-\rho} + \delta \right) \right] = \max_{0 \le \rho \le 1} \underbrace{\left[\rho^r \frac{1-\rho}{2-\rho} \right]}_{\delta_{\mathrm{TG}}(\rho)}. \tag{119}$$

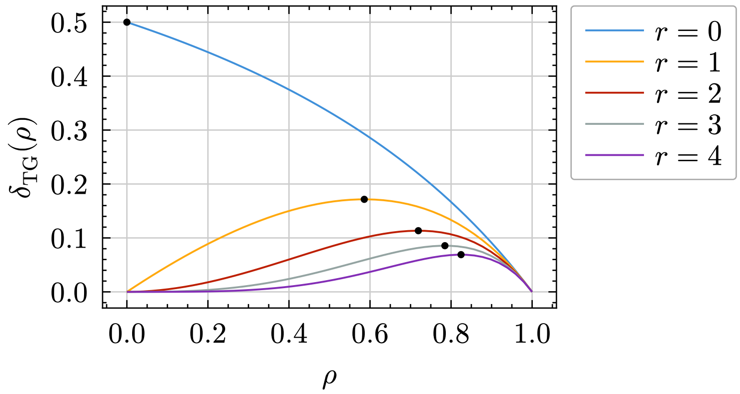


Figure 16: Maxima of $\delta_{\text{TG}}(\rho)$ for different values of r.

The maxima of $\rho_{\text{TG}}(\rho)$ can be found analytically by setting its derivative to zero. However, the resulting term for the maximum is rather complicated. Braess found a good upper bound:

$$\rho_{\text{TG}} < \frac{1}{(r+1)e}. \tag{120}$$

11. The General Multigrid Method

When we have more than two levels, $\delta = (\delta_{q-1})^{\mu}$ for $q > 1$ (Equation 93). Then the norm reduction becomes

$$\delta_q = \max_{0 \le \rho \le 1} \left[\rho^r \left(\left(1 - (\delta_{q-1})^{\mu}\right) \frac{1-\rho}{2-\rho} + (\delta_{q-1})^{\mu} \right) \right] \tag{121}$$

Note that this recursive function reflects the recursive nature of the multigrid algorithm. We assume that the multigrid convergence is independent of the level q, i.e.,

$$\delta_q = \delta_{q-1} =: \delta_{\text{MG}}. \tag{122}$$

Then we find

$$\delta_{\text{MG}} = \max_{0 \le \rho \le 1} \left[\rho^r \left(\left(1 - (\delta_{\text{MG}})^{\mu}\right) \frac{1-\rho}{2-\rho} + (\delta_{\text{MG}})^{\mu} \right) \right] \tag{123}$$

This equation can be solved numerically. The resulting convergence rates can be seen in Table 2.

$r \rightarrow$	0	1	2	3
Two-Grid	0.5	0.172	0.114	0.086
MG $\mu = 1$ (V-cycle)		$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$
MG $\mu = 2$ (W-cycle)		0.187	0.120	0.087

Table 2: Numerical values of the convergence rate for the two-grid method and the general multigrid method, as given by Braess.

Note that these convergence rates are far below 1, proving not only that the multigrid method converges but also that it converges *fast*.

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